

■ ReplaceAll

$expr /. rules$ applies a rule or list of rules in an attempt to transform each subpart of an expression $expr$.

Example: $x + 2 /. x \rightarrow a \longrightarrow 2 + a$. ■ `ReplaceAll[expr, rules]` is equivalent to `MapAll[Replace[#, rules], expr]`. ■ `ReplaceAll` looks at each part of $expr$, tries all the $rules$ on it, and then goes on to the next part of $expr$. The first rule that applies to a particular part is used; no further rules are tried on that part, or on any of its subparts. ■ `ReplaceAll` applies a particular rule only once to an expression. ■ Example:
 $x /. x \rightarrow x + 1 \longrightarrow 1 + x$. ■ See the notes on `Replace` for a description of how rules are applied to each part of $expr$. ■ $expr /. rules$ returns $expr$ if none of the rules apply. ■ See page 199. ■ See also: `Rule`, `Set`.